

based emerging technologies in the market. Some of the listed emerging technologies will be dropped after due course of time if they do not contribute significantly as per expectations and market demand. The chart can also be helpful for innovators and developers to work on these technologies.

## AI startups

As per reports from *Forbes.com*, over 154,000 AI patents have been filed worldwide since 2010. Some of these companies are getting positive results and their work has paid off well. AI technology has great potential across many industries, from retail, manufacturing to energy and healthcare.

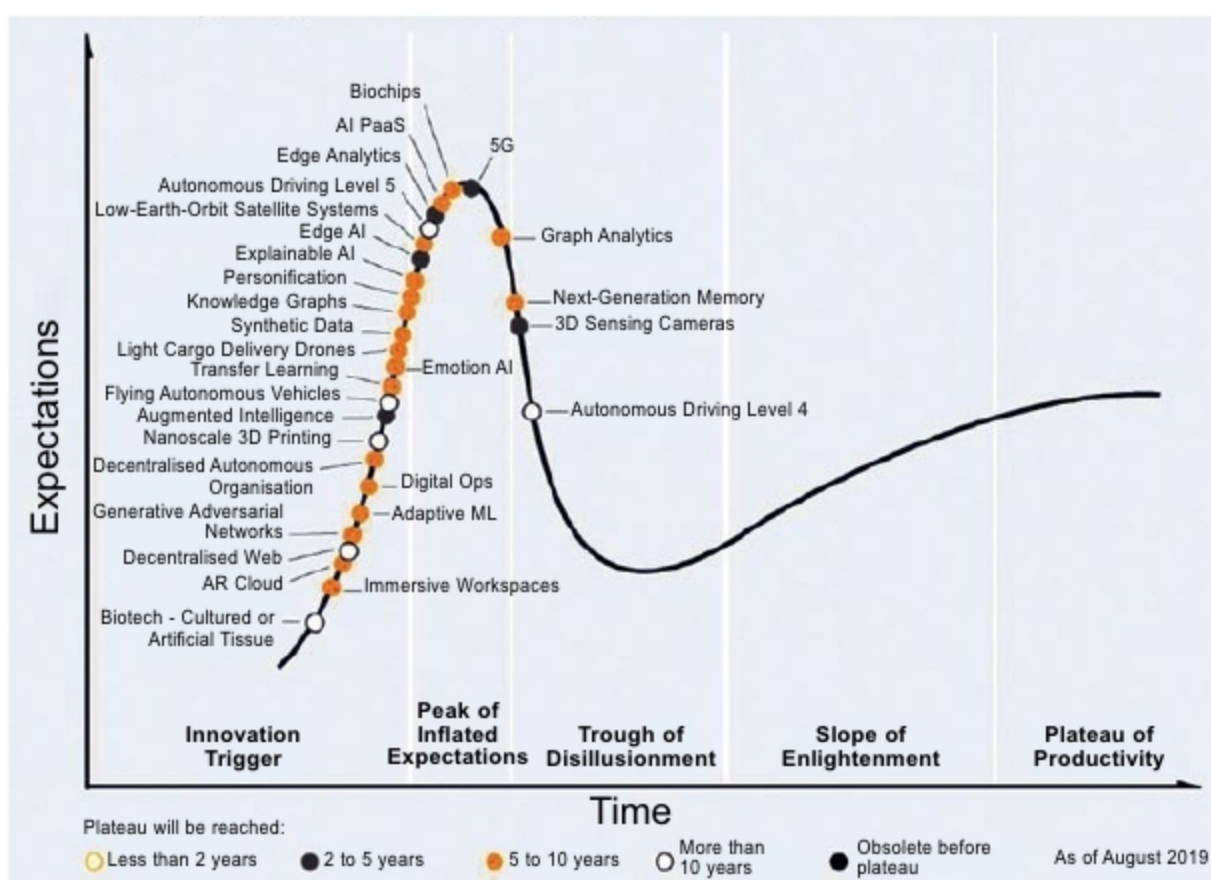
According to research firm PricewaterhouseCoopers, in 2018, billions of dollars were invested in AI companies in Europe, and the industry is projected to be worth more than fifteen trillion dollars by 2030. But not all companies that claim to use AI are telling the truth. Many are making vague claims of using AI in their products and services to secure funding or attract customers. A recent study by London-based venture capital firm MMC found that, out of 2830 European startups classified as AI companies, only 1580 accurately fit the description. It was found that forty per cent of AI startups in Europe do not actually use any AI technology.

While it is true that Indian companies are following AI technology trends, there are also reports that many of them are not working on leveraging AI technology. These so-called AI-based companies, which have received millions in funding from various sources, are reportedly relying mostly on human work force, while using the hype around AI to attract customers and investments. India has started to scale its tech entrepreneur presence in global markets but is still way behind many countries including the US, China and Israel in AI technology.

## AI and robots

People often confuse AI with robots. Robots are programmable machines that are used to carry out a series of actions manually, semi-autonomously or autonomously. Many of them are not intelligent.

Whereas, AI is a branch of computer science that involves developing



Gartner's hype cycle (Credit: gartner.com)

computer programs. For example, AI algorithms are used in Google searches, Amazon's recommendation engine, Siri, Alexa and others. Hence, not all AI programs are used to control robots. Artificially-intelligent robots, or AI robots, are a combination of robotics and AI. These robots are controlled by AI programs.

## AI without machine learning

We have come across many articles that say there is no AI without machine learning. It is also said that AI without machine learning is like a car with no engine. Is AI without machine learning possible?

Yes, it is! The heart of AI is not always machine learning. Researchers in early days found ways of creating AI without machine learning. These AI techniques are still alive and used even today.

Initially, machines were designed to formulate outputs based on inputs represented by symbols. Symbolic AI research was popular from the mid-1950s until the late-1980s. Good old-fashioned AI, or GOF AI, was based on a human-understandable symbolic system.

GOF AI was first used by John Haugeland in his 1985 book titled *Artificial Intelligence: The Very Idea*. It is AI without machine learning. So, behind all the hype and excitement of machine learning, there is GOF AI

in the background. For example, one common use of GOF AI in today's applications is the chatbot. Many chatbots are solely based on GOF AI, despite the immense progress of natural language processing through machine learning.

Symbols are used when input falls under certain and definite conditions. However, when there is uncertainty involved, like predictions, representation is done using fuzzy logic as seen in artificial neural networks. As we go deeper into AI, we find that machine learning is a key element in AI technology. It is mostly based on using lots of training data and good algorithms. There is also a possibility that machine learning can exist without good algorithms, but not without good data.

## Individualism social phenomena

AI technology has benefited not only business but the life of every individual. AI-enabled smartphones and robots have had a huge impact on the modern society. But some experts feel that these emerging technologies also create many problems. With advancements in such technologies as mobile and AI, society is becoming something of the past. Human species live together in a society in one way or the other. Society may curb an individual's independence but it also provides many benefits including valuable companionship.